YASHWANT RAJ

Data Scientist | Machine Learning Engineer | AI Developer

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PROFESSIONAL SUMMARY

Results-driven 4th-year Data Science and Machine Learning student experienced in building and deploying end-to-end AI/ML systems across deep learning, computer vision, and generative AI. Skilled across the full pipeline—from data preprocessing to model training and production deployment—using TensorFlow, Keras, OpenCV, and LangChain. Seeking a Data Scientist or Machine Learning Engineer internship.

EDUCATION

BSc (Hons) in Data Science and Machine Learning — RV University, Bangalore *2022 – 2026 | GPA: 8.7 / 10*

TECHNICAL SKILLS

Languages: Python, SQL, C
Frameworks: TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, LangChain, Matplotlib
AI / ML Domains: Computer Vision, Generative AI, NLP, Predictive Modeling, Deep Learning, RAG Systems
Tools & Deployment: FastAPI, FAISS, Ollama, Docker, Power BI, Tableau, MySQL, SQLite, Streamlit, Flask, Raspberry Pi (IoT), Git, Excel
Soft Skills: Problem Solving, Communication, Presentation, Leadership, Team Collaboration

EXPERIENCE

Data Science Intern — Dhee Center for AI and Data Science, RV University *June 2024*

- Engineered a real-time facial emotion detection system using CNNs and OpenCV, achieving 92% classification accuracy across 7 emotion classes on diverse demographic datasets.
- Implemented real-time facial emotion recognition from live video, enabling continuous frame-by-frame inference at 24+ FPS and reducing manual monitoring effort by ~60%.
- Performed end-to-end preprocessing, EDA, and model evaluation using k-fold cross-validation; improved robustness across varied lighting conditions, reducing false positive rate by 18%.
- Benchmarked CNN performance against SVM and ResNet baselines; authored technical documentation and identified deployment use cases in healthcare, HR, customer service, and entertainment.
- Collaborated with a cross-functional team of 4 engineers; managed source code/model versioning using Git, following Agile practices to deliver sprint milestones on schedule.

PROJECTS

AI-Based Driver Drowsiness Detection System | Computer Vision + IoT Deployment

- Built a real-time drowsiness detection system achieving 96% detection accuracy on a 2,000+ image dataset, improving driver safety monitoring for automotive and fleet management applications.
- Trained a TensorFlow/Keras CNN to classify alert vs. drowsy states; applied OpenCV face detection, eye tracking, and Eye Aspect Ratio (EAR) for continuous analysis.
- Deployed the model on a Raspberry Pi IoT device for edge-based, privacy-preserving, low-latency inference with a real-time buzzer/alarm alert system optimized for 512 MB RAM.
- Designed a modular, version-controlled (Git) pipeline covering data collection, preprocessing, training, inference, and alert generation, ensuring scalability for future hardware upgrades.

Technologies: *Python, TensorFlow, Keras, OpenCV, Raspberry Pi, IoT, NumPy, Git*

Enterprise RAG-Based Customer Support Chatbot | Retrieval-Augmented Generation System

- Designed and developed a scalable Retrieval-Augmented Generation (RAG) chatbot answering domain-specific queries across 500+ pages of company documentation, cutting manual query resolution time by ~70%.
- Built a modular ingestion pipeline supporting 3 document formats (PDF, DOCX, TXT) with automated preprocessing and metadata extraction, processing 1,000+ documents into 10,000+ indexed chunks.
- Implemented semantic chunking and Sentence Transformer embeddings feeding a FAISS vector database, enabling sub-second similarity search and improving retrieval relevance by ~30% over keyword search.
- Built a top-k retrieval and context-ranking pipeline that injects relevant chunks into a locally-hosted LLM (Ollama), enabling secure, offline inference with zero reliance on external APIs.
- Designed a modular FastAPI backend with 6+ independently scalable services and persistent SQLAlchemy/SQLite storage; containerized with Docker and architected for future AWS EC2 migration.

Technologies: *Python, FastAPI, LangChain, FAISS, Sentence Transformers, Ollama (Local LLM), SQLAlchemy, SQLite, Docker, AWS (Planned), Git*

CERTIFICATIONS

- Deep Learning for Developers — Infosys Springboard
- Introduction to Machine Learning — Infosys Springboard